

The complete method

The full body of work behind a single-strategy platform: every approach explored, validated, or rejected, and the discipline that decided. Twenty-plus strategies, one honest scorecard, and the reason only a few of them will ship.

by **Carter Warrens** · 2026 · reference implementation in Julia · figures are historical backtests and paper results on ETF and equity data

THE PROGRAM IN ONE PARAGRAPH

Blaque Baux began as an alpha-prediction system and became a risk-premium harvester. The reason: rigorous out-of-sample testing found no exploitable predictive edge at the horizons and instruments we can access. We reproduced that null result a dozen ways. Cross-sectional factors, Bayesian filtering, 15-minute signals, cross-industry lead-lag, and pairs arbitrage were all coin flips. What *survives* is **risk structure**: diversification and trend premia, harvested by a governed daily process. Around that core we ran a deep second program on **convexity and tail protection**, on **correlation structure**, and on **aggression and leverage**, each ending in a durable, often counter-intuitive law. This paper is the map of all of it: what shipped, what is in live testing, what was kept as research, and, just as prominently, what was tested and rejected, with the reason. The honesty is the point. A strategy is only as trustworthy as the ideas it was willing to kill.

1 The founding finding

On 60-day 15-minute data (about 1,547 bars across 60 S&P names), two independent alpha models scored **50.9%** and **50.7%** directional accuracy. Both were coin flips, with an information coefficient near zero. We then reproduced the null across blue-chip and mid-cap universes, cross-industry lead-lag, and pairs mean-reversion. The conclusion was not "try harder"; it was **structural**. At retail-accessible horizons and costs, the market has already priced what is knowable. So the platform stopped forecasting returns and started harvesting the premia paid for *bearing risk* (diversification, trend, term and asset premia), where an out-of-sample edge genuinely survives.

2 The scorecard

As shown below is every strategy and idea explored, with its verdict and status. Green is live or graduating, amber is kept as research, red is tested and rejected. This is the whole program at a glance.

STRATEGY / IDEA	CATEGORY	FINDING	STATUS
LIVE & PRODUCTION			
The Spine (inverse-vol/ERC base + 12-mo TSMOM trend + vol-target + regime brake)	Risk-premium harvest	Sharpe ≈ 0.97 , maxDD $\approx -11\%$, 2022 -1.4% . The trend sleeve carries crises. Forecasts nothing.	LIVE
Agriculture sleeve (DBA, the DB Agriculture Fund) as a 6th holding	Diversification	The only genuinely uncorrelated add tested (0.44 correlation to broad commodities). Sharpe 0.94 $\rightarrow$ 1.04.	LIVE
GRADUATED: LIVE PAPER A/B (THREE ISOLATED ACCOUNTS)			
Multi-horizon trend (3/6/12-mo agreement)	Funded convexity	Strict improvement: Sharpe 1.04 $\rightarrow$ 1.16, drawdown $-8.9\% \rightarrow -7.0\%$, more convex, no cost.	A/B
Split-universe spine (base and trend on different universes)	Breadth	Trend on a broad shortable CTA set gives Sharpe ≈ 1.12 at lower vol. Substitutes with multi-horizon.	A/B
REVIEWED, KEPT AS RESEARCH (NOT DEPLOYED)			
Gamma-ARMA regime framework (Bayesian regime over ARMA/GARCH space)	Regime / risk	Sophisticated and specified, but <i>unvalidated</i> (its backtest was never run). Path-A funded research.	RESEARCH
Taleb barbell (90% safe / 10% convex)	Convexity	Paid insurance that works: skew +2.1, positive in every crisis (+19% in the COVID crash while equities fell 33%), at a bleed of about 3%/yr.	RESEARCH
The "curveball" (reversed barbell, a 10/90 convex sliver)	Convexity	A dark horse: ruin if compounded ($-44\%/yr$) but +237% in the COVID crash on its 10% weight. Hold it, let it bleed, never time it.	RESEARCH
Inverse carry (short high-carry, i.e. short the premium)	Tail hedge	A positive-skew hedge (skew flips -2.1 to $+2.1$) that catches the rate/credit crisis (2022 +5%) VIXY misses.	RESEARCH
Diversified tail hedge (across crisis regimes)	Convexity	Deflation and inflation hedges are anti-correlated (-0.3), so a blended sliver is positive in <i>every</i> crisis.	RESEARCH
Optimizer library (Markowitz, Black-Litterman, CVaR/CDaR, HRP, Michaud, cost-aware)	Infrastructure	A general-purpose portfolio-optimization stack plus REST service. The spine is one strategy built on it.	TOOL
TESTED & REJECTED: RETURN PREDICTION (ALPHA)			

STRATEGY / IDEA	CATEGORY	FINDING	STATUS
3-factor cross-sectional & Bayesian (Kalman/NIG/James-Stein/ADVI) alpha	Alpha	50.9% and 50.7% directional accuracy, IC $\approx$ 0. Coin flips. Archived.	REJECTED
Blue-chip & mid/small-cap directional prediction	Alpha	Blue-chip 12-mo momentum 52.1% (IC-IR 1.78, below the 2.0 significance bar); mid-cap momentum negative at 48.3%, worse than a coin.	REJECTED
15-minute intraday alpha	Alpha	50.9% accuracy, noise. Intraday cadence adds nothing, since diversification is the real shock absorber.	REJECTED
Lead-lag / earnings read-through (NVDA $\rightarrow$ semis, banks, tech)	Alpha	Priced same-day (corr 0.7 to 0.9), next-day $\approx$ 0. In tech it reverses: chasing a MSFT/GOOG pop returned -1.1% vs a +0.4% baseline.	REJECTED
Pairs / relative-value mean-reversion (stat-arb)	Alpha	Diversified book $\approx$ 0.15 Sharpe after costs, i.e. noise. Many "pairs" drift apart permanently (KO/PEP reversion half-life 3,800 days).	REJECTED
TESTED & REJECTED: CONVEXITY, CARRY, LEVERAGE			
Vol-aware VIXY tail overlay on the Spine	Paid convexity	Hurts: cut CAGR from 5.9% to 1.5% and deepened drawdown from -8.9% to -18.3%. The Spine's trend is already a crisis hedge; VIXY carry is a drawdown machine.	REJECTED
Convex response function (size by trend strength ²)	Convexity	Backfires: flattened sleeve skew from +0.47 to -0.08 and deepened drawdown to -19.5%. Convexity must live in the payoff, not the sizing.	REJECTED
Carry sleeve as a third premium	Risk premium	Genuinely uncorrelated (-0.02 to base) but negative-expectancy with negative skew (-2.5), so it lowered the blended Sharpe from 1.16 to 0.96.	REJECTED
Carry as the barbell's earning sleeve	Convexity	Crashes when the convex sleeve pays, eating the payoff: COVID capture shrank from +17% to +6% and 2022 turned from +2% to -3%.	REJECTED
Leverage to double-digit returns	Aggression	Net-negative at retail financing: 3 $\times$ cuts CAGR to 3.9% at a -38% drawdown. Reaching double-digit CAGR needs about 3 $\times$, i.e. a -39% drawdown. Kelly $\approx$ 1 $\times$.	REJECTED

3 What's live, and what's graduating

Production: the Spine, a two-sleeve book (inverse-vol/ERC base plus 12-month time-series-momentum trend). Each sleeve is volatility-targeted on its own realized P&L, blended 50/50, and

scaled by a drawdown regime brake, then rebalanced once daily and routed through a governed execution layer. Net of costs it delivers a Sharpe near 1.0 at about 11% maximum drawdown, with the trend sleeve making money in 2022 while a long book bled. It now trades six assets (SPY, IEF, TLT, GLD, DBC, DBA).

In live A/B (three isolated paper accounts): single-sign (the control), split-universe, and single-multi-horizon. The two candidate upgrades are measured head-to-head against production on real fills, each with its own state, ledger, and safety files. Nothing graduates to funded capital until it clears out-of-sample and purged cross-validation.

4 The research themes

Return prediction: the null result, reproduced

Every route to forecasting returns landed in the same place. The two original alpha engines scored **50.9%** and **50.7%** directional accuracy on 15-minute data, a coin flip with information coefficient near zero. Blue-chip 12-month momentum managed only **52.1%** with a t-statistic of **1.78**, below the significance bar of 2.0; mid-cap momentum was outright negative at **48.3%**. Cross-industry read-through was priced the *same day* (correlation 0.7 to 0.9) with next-day predictability near zero, and negative in tech, where chasing a Microsoft or Google earnings pop returned **-1.1%** against a +0.4% baseline. A diversified pairs-arbitrage book netted **0.15 Sharpe** after costs, indistinguishable from noise. Five independent methods, one answer. That consistency is the real evidence: at horizons and costs a retail participant can reach, the market has already priced what is knowable, and it prices it instantly. So we stopped forecasting and built around the risk premia that survive.

Convexity and tail hedging: free vs paid

The deepest thread. Convexity comes in two forms. **Free** convexity (trend and momentum, funded by the momentum premium) protects sustained crises and improves return. **Paid** convexity (long volatility or short carry, funded by decay) protects sharp crashes but bleeds and cannot be timed. A paid overlay *hurts* the Spine, which already owns free convexity; making the trend sleeve multi-horizon is the win instead. The best standalone hedge is diversified across crisis regimes: deflation (VIXY, short-credit) and inflation (short-duration, long-commodities) are anti-correlated, so a small sliver has no blind spot.

Correlation structure: real, but priced

Industries, supply chains, and firms are strongly correlated (sectors about 0.62, big banks about 0.87), but the read-through is *instantaneous*, which makes it a risk tool rather than an alpha source. And correlation is proportional to **shared forced exposure**: prudential regulation (banks, utilities) or a common input (airlines share one fuel) homogenizes firms into one factor, while product regulation with idiosyncratic outcomes (pharma and biotech under the FDA) leaves them uncorrelated despite heavy oversight.

Aggression and leverage: the growth-vs-ruin frontier

"Most aggressive" has a principled ceiling (Kelly) and a brutal catch: aggression multiplies edge, and ours is modest. At retail financing, leverage *lowers* growth; a max-aggressive convex bet has a median terminal wealth below the stake. Rational aggression is fractional Kelly on real edge, or a small barbell sleeve, never all-in.

The archived alpha track

The original Bayesian alpha engine, a Monte-Carlo risk simulator, a Sharpe-on-historical-means "Oracle," and the Gamma-ARMA regime framework are preserved but archived. The mathematics is sound; the measured edge was not there. Sophisticated machinery *packages and sizes* a signal. It does not *create* one.

5 The durable laws

The findings that survive the small-sample caveats. Directional and structural, they are the real output of the program.

I. At tradeable horizons there is no exploitable return-prediction edge, so harvest premia rather than forecast.

II. Convexity is either *free* (trend, for sustained crises) or *paid* (long-vol, for sharp crises). No instrument gives cheap, timed, crash-day protection.

III. Timing the tail removes the tail. Every rule that trims a hedge's bleed also removes its payoff.

IV. Diversification is necessary but not sufficient. An uncorrelated stream only helps if it is positive-expectancy.

V. A hedge's safe sleeve must be inert in a crisis. Anything with hidden crisis-beta is not safe.

VI. There are two crisis regimes, deflation and inflation. They are anti-correlated and need opposite hedges.

VII. Aggression multiplies edge. With none, it multiplies only ruin (Kelly is about 1× here).

VIII. Correlation is real and priced instantly. It is a risk tool, not an alpha source.

IX. Correlation is proportional to shared forced exposure (prudential regulation or a common input); idiosyncratic outcomes decorrelate.

X. The discipline, validate out-of-sample then deploy, is the actual product. A strategy is only as trustworthy as the ideas it killed.

6 Governance: the discipline is the product

Every order passes one governed path enforcing hard, tested invariants: idempotency, per-pool budget and daily-loss limits, data-staleness, position reconciliation, a kill switch, and full fill lineage, all beneath a pre-trade safety gate (drawdown and loss halts, gross-leverage and per-name caps, account and data checks). No language model is anywhere near the order path; the strategy is reproducible code. The autonomous daily job runs deterministically pre-open with morning catch-up fires, and reports its book and P&L to a comparison log. Nothing reaches capital without clearing the same out-of-sample bar the Spine did, which is why most of the scorecard is red. That is a feature.

7 Honest caveats

Backtests are historical or paper results on ETF and equity proxies, net of modeled costs, and are not indicative of future results. Tail and earnings findings are *especially* sample-dependent, because the window holds a single sharp crash (COVID) and a single inflation bear (2022). Several positive research results (inverse carry, short-duration hedges) lean on that 2022 window and should be read as *shapes*, not promises. The validated core is single-digit-CAGR, roughly 1.0-Sharpe money. There is no double-digit-return-at-low-risk configuration, and this program says so plainly.

Companion documents. Validated methods and math: [docs/whitepaper.html](#), [docs/FINANCIAL_METHODS.md](#). Research notes: [docs/research_thread_summary.pdf](#), [docs/barbell_note.pdf](#), [docs/research_frontier.html](#). Decisions: [docs/CANONICAL_ARCHITECTURE.md](#). Reproducible sketches: [scripts/research/](#) (11 scripts). Reference implementation in Julia: the Blaque Baux repository.

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